
Robo Rapture (Audio)

Proposition by Brandon and Fontaine



Meet the Audio Leads !

Brandon Corona

Team Lead for:
Audio Programming
Music System Design



Fontaine Chu

Team Lead for:
Sound Design
Reaper Implementation

Description



RoboRapture is a 2.5D game where we follow **Brutus**, **Maximilian** and **Alessa**, a group of robots in a journey to kill **The Rapture**. Throughout a diverse set of biomes, the robots will face different demonic enemies and hellish bosses to **achieve victory**.



The player, through positional and tactical play; will try to kill all enemies unlocking special abilities and survive through the end.



Theme

Our Game Audio approach will achieve a dynamic gameplay of a dark, foreboding world, and all the sound designs which reflect that.

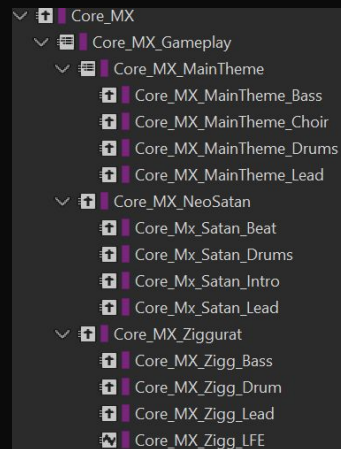
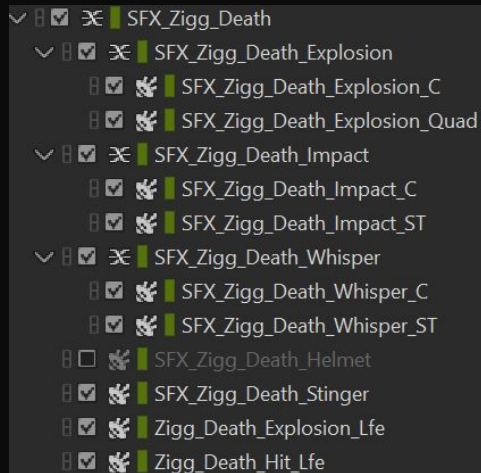


Being the gory, fleshy enemies and creepy ambiances, or the uplifting and epic attacks; the player will engage in a **glorious battle for survival.**



We only used Mono, Stereo, Quad, and LFE to have more control over the audio. We avoided silence at the beginning and used delay to save space.

We also have different stems for our music and separate SFX into different layers to enable more detailed and creative side-chain processing



Our Workflow

Field Recording/ADR Recording

Sound Design in a DAW

Pro Tools

Reaper

Bounced

Using Reaper Extension from
Audiokinetic - ReaWwise

Imported into Wwise

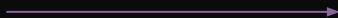


Sound Design

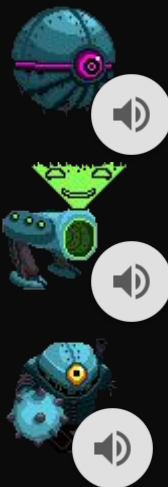
- **SIZE:** Sounds used a different channel configuration depending on their importance
- **MOVEMENT:** Sounds are moving from inside to outside the screen and vice versa
- **DETAIL:** Coverage of even the tiniest of details and as clear as possible to denote sound.



Sound Design Example



Hit SFX



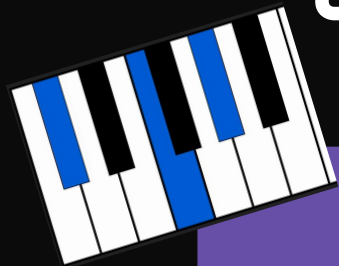
Music



- **EXCITEMENT:** To make the game as exciting as possible, due to being constantly a similar gameplay
- **STEM VARIATION:** Because the music is constantly playing, the stems will vary according to the SFX sidechain system
- **DYNAMIC:** Different stages of the game means different music, as well as pause menus, main menus, etc.

Features

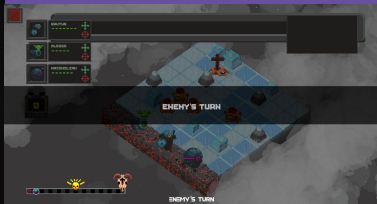
Core Audio Managers



WWISE MENU
MANAGER



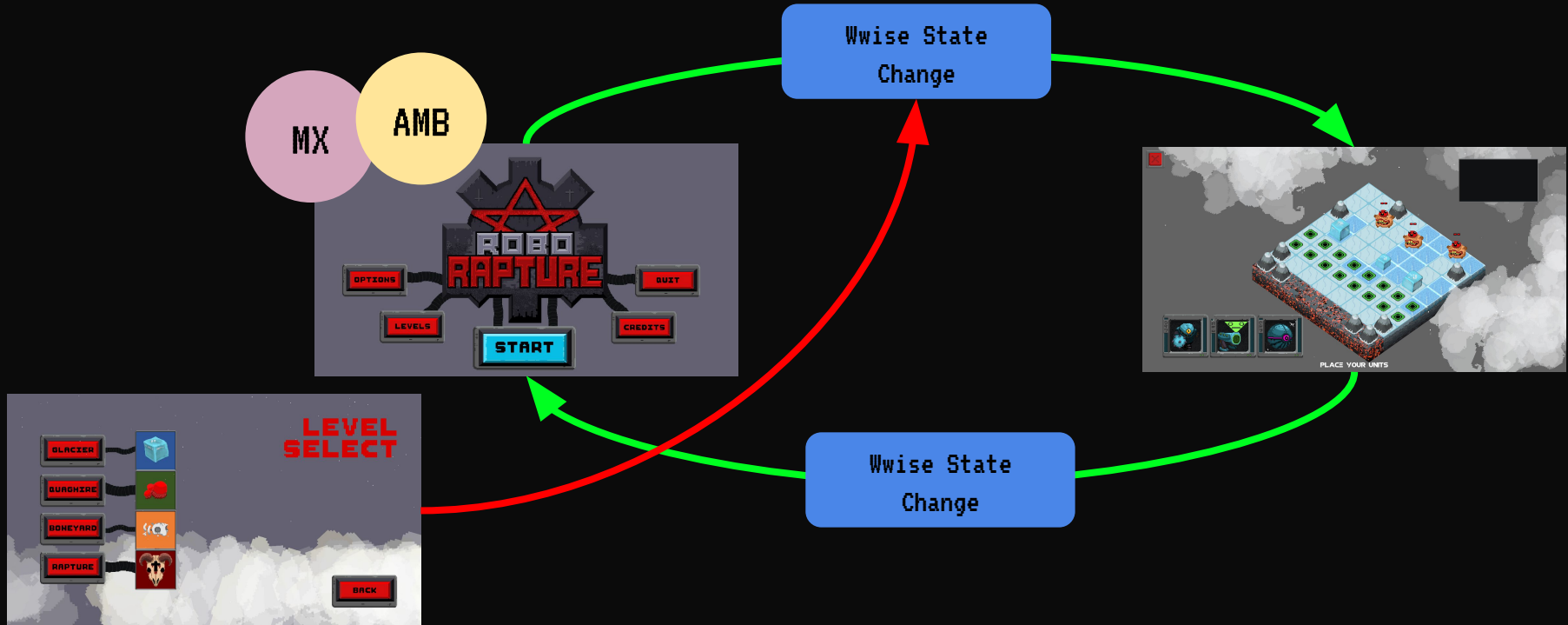
WWISE TURN
MANAGER



WWISE
DIALOGUE
SYSTEM

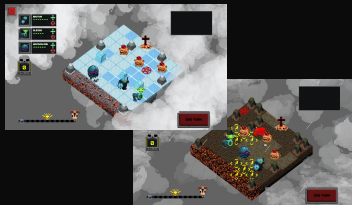


Menu Manager



Turn Manager

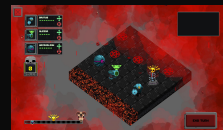
Glacier/Quagmire



Boneyard



Rapture



AMB

Wwise State
Change

Wwise State
Change

Wwise State
Change

MX

Wwise State
Change

Wwise State
Change

Wwise State
Change



Character Selection

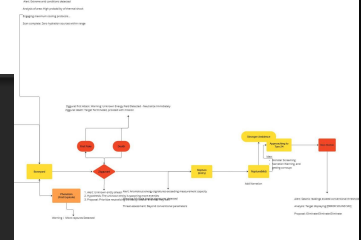
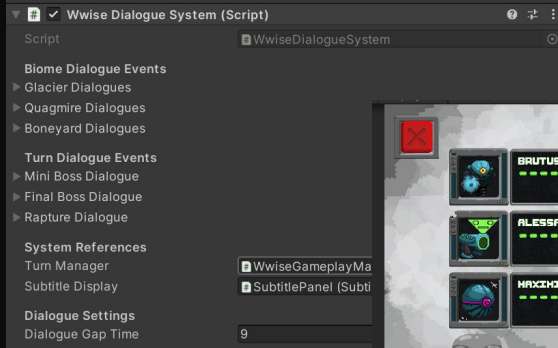


Mini Boss



Final Boss

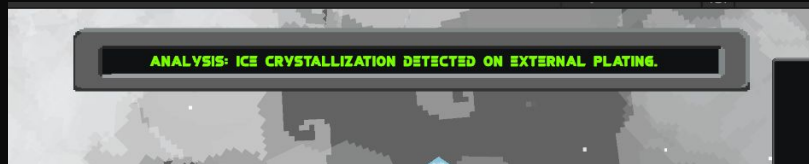
Dialogue System



Player Help

Main Concept

- Inspired by Nier:Automata's Pod System
- Functions as a dynamic player guidances system

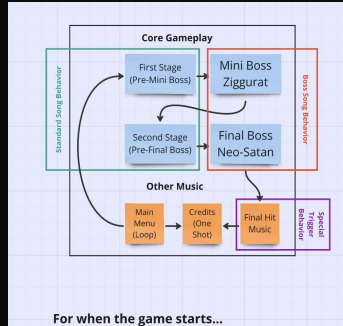


When will it trigger?

- Upon entering new environments
- When approaching boss encounters

ORGANIZATION

Documentation & Organization

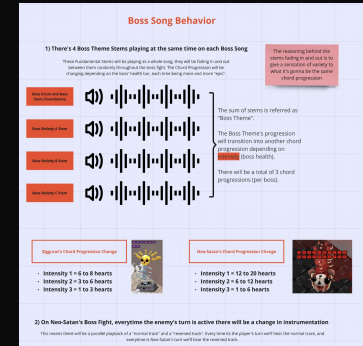
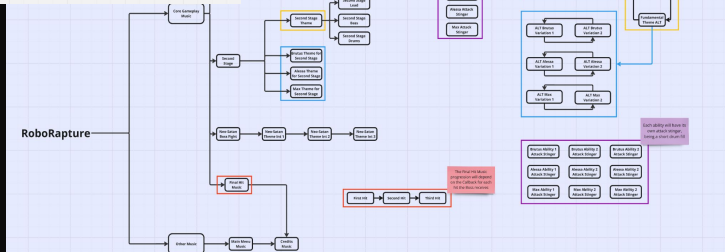
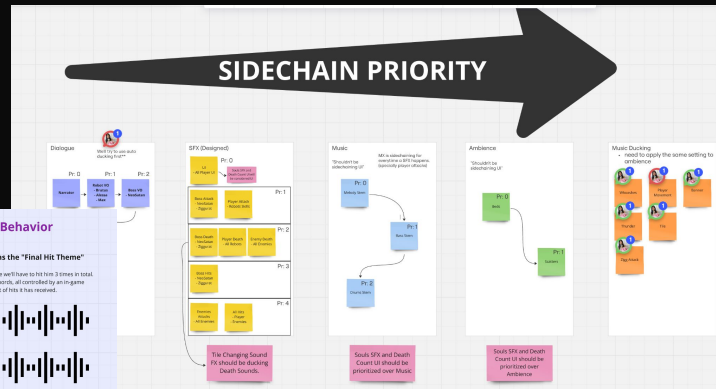
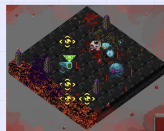


- The fundamental Bass will be muted until the player spawns all characters. Also there will be a low pass filter on the Main Theme around 1K that will disappear linearly (and quickly) after the last character appears.
- The Character Stems will do an epic arpeggi to create a sense of needing to resolve into the fundamental note, after all characters are placed then we will hear the resolution.



There's one single stem playing as the "Final Hit Theme"

After we beat Neo-Satan, in order to beat the game we'll have to hit him 3 times in total. Every time we hit him, we'll hear a change in chords, all controlled by an in-game



Asset List

Asset Count																
		RoboRapture					Brandon					Fontaine				
		SFX	UI	MX	AMB	VO	SFX	UI	MX	AMB	VO	SFX	UI	MX	AMB	VO
Asset Creation	Total	119	33	24	12	32	72	32	24	1	0	47	1	0	11	32
	Not Started	1	0	0	1	0	0	0	0	1	0	1	0	0	0	0
	In Progress	0	0	24	11	0	0	0	24	0	0	0	0	0	11	0
	Designed	118	33	0	0	32	72	32	0	0	0	46	1	0	0	32
Wwise/Unity	Assigned	N/A					1	0	24	0	0	118	33	0	12	32
	Implemented	118	33	24	9	32	1	0	24	0	0	117	33	0	9	32
Creation Totals	Done / To do	183 / 220					104 / 129					79 / 91				
	Percentage	83.2%					80.6%					86.8%				
Wwise/Unity Totals	Done / To do	216 / 220					25 / 25					191 / 195				
	Percentage	98.2%					100.0%					97.9%				
Grand Total		90.7%					90.3%					92.4%				

Red for < 30%
Yellow for < 60%
Green for < 90%
Blue for > 90%

Game Syncs

	SFX	UI	MX	AMB	VO
Total	10	0	22	12	0
Not Started	0	0	0	0	0
In Progress	0	0	7	1	0
Designed	10	0	15	11	0

```
=COUNTIFS(UI!H:H, "Brandon", UI!I:I, "Not Started ")
```

```
=SUM(I9:M9) & " / " & SUM(I6:M6)
```

```
=COUNTIF(AMB!N:N, "In Progress") + COUNTIF(AMB!N:N, "Implemented")
```

CASE STUDY

Problem

- We want to rename all our assets

Goal

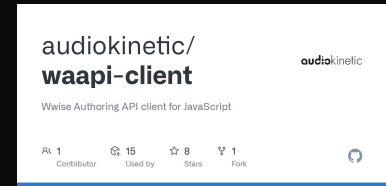
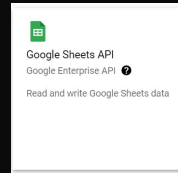
- Speed up workflow
- Create a new tool

Challenge

- There are more than **200** items in google sheet and more than **1700** items in wwise need to be renamed

Result

- Python Based Renaming Tools (Wwise and Google Sheet)



CASE STUDY #2 - Naming Convention Issue

Problem

- Playing through the game at normal speed takes at least 15 minutes to complete.

Challenge

- The technical sound designer in charge of
 - Check for missing sounds
 - Ensure ambience and dialogue trigger appropriately
- It takes a lot of **TIME**

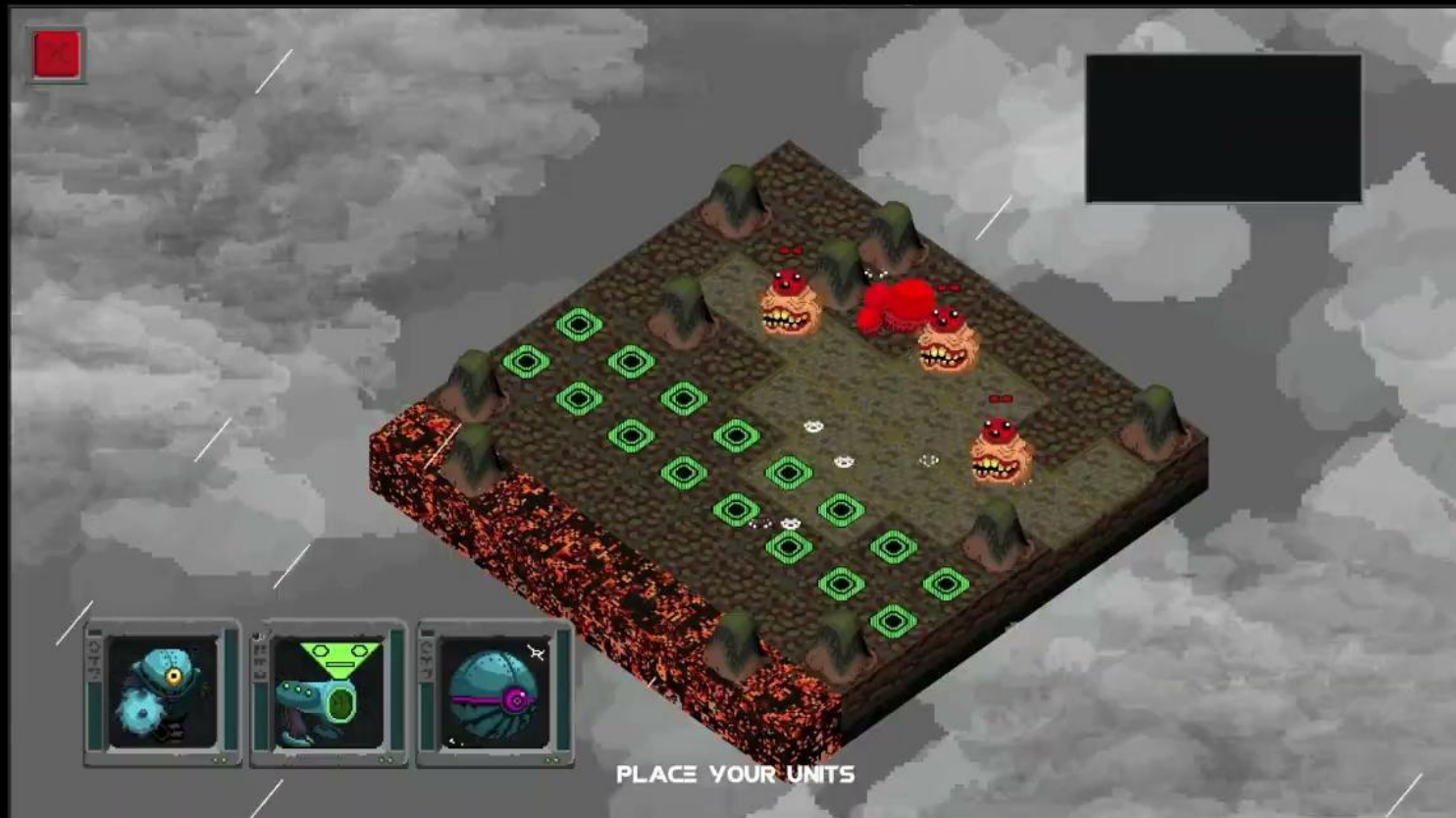
Goal

- Speed up workflow
- Create a new tool

Result

- Inspired by Hokai Star Rail Speed up functions
- Helps reduce time spent in repetitive fight





Challenge

- Reduced Team Size
- We have multiple projects (personal work and group work need to work on)

Actions Taken

- **Communication**
- Hard-Working
- Believe in yourself!



Goal

- Finish all works on time

Result

- **Successfully completed project, maintaining project scope and quality**

THANKYOU

